

Subject:
Organic Chemistry

System:
Introduction to Organic Chemistry

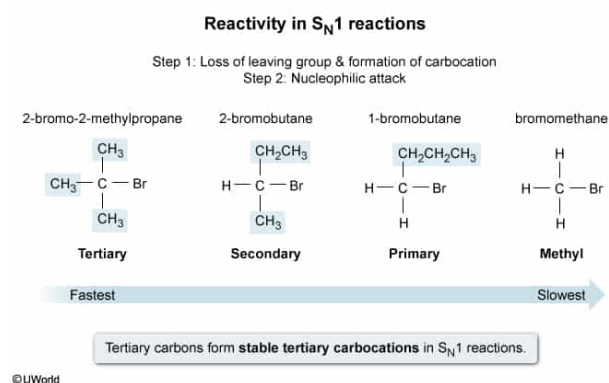
Four alkyl halides: bromomethane, 2-bromobutane, 2-bromo-2-methylpropane, and 1-bromobutane, each react with water through solvolysis by an S_N1 reaction in which the solvent acts as the nucleophile. In what order will these alkyl halides react to form alcohols, from fastest to slowest?

- ☐ A. bromomethane $\rightarrow$ 1-bromobutane $\rightarrow$ 2-bromobutane $\rightarrow$ 2-bromo-2-methylpropane [21%]
- ☐ B. bromomethane $\rightarrow$ 1-bromobutane $\rightarrow$ 2-bromo-2-methylpropane $\rightarrow$ 2-bromobutane [2%]
- ✓ ☒ C. 2-bromo-2-methylpropane $\rightarrow$ 2-bromobutane $\rightarrow$ 1-bromobutane $\rightarrow$ bromomethane [73%]
- ☐ D. 2-bromobutane $\rightarrow$ 2-bromo-2-methylpropane $\rightarrow$ 1-bromobutane $\rightarrow$ bromomethane [3%]

Correct

73% answered correctly

Explanation:



S_N1 reactions are nucleophilic substitutions that occur in **two steps**. During the first step, the bond between the leaving group and carbon is broken and a **carbocation** intermediate is formed. A nucleophile attacks the carbocation in the second step, forming a bond between the nucleophile and the carbocation to yield the final product.

Reactivity of the alkyl halide substrate is dependent on its level of **substitution**.

Tertiary alkyl halides form the most **stable carbocations** and undergo S_N1 reactions **more readily** than less-substituted alkyl halides. As alkyl halide **substitution decreases**, the **carbocations** become **less stable** and the reaction slows.

The substitution of the four alkyl halides 1-bromobutane, 2-bromobutane, 2-bromo-2-methylpropane, and bromomethane are primary, secondary, tertiary, and methyl, respectively. In the **solvolysis reaction** with H_2O , the tertiary alkyl halide 2-bromo-2-

methylpropane will react the fastest because it forms the most stable carbocation in the first step. The secondary alkyl halide will react second fastest, followed by the primary and then the methyl alkyl halides. Therefore, the order of reactivity from fastest to slowest in an S_N1 reaction is **2-bromo-2-methylpropane** $\rightarrow$ **2-bromobutane** $\rightarrow$ **1-bromobutane** $\rightarrow$ **bromomethane**.

(Choice A) The alkyl halides here are arranged from *slowest* to *fastest* reactivity in S_N1 reactions.

(Choice B) Bromomethane (methyl halide) forms the least stable carbocation, and therefore reacts slowest in S_N1 reactions. The tertiary and secondary halides, 2-bromo-2-methylpropane and 2-bromobutane, form stable carbocations and should be listed before the primary and methyl halides.

(Choice D) The order of the first two alkyl halides is switched. The tertiary alkyl halide, 2-bromo-2-methylpropane, forms a more stable carbocation and will react faster than the secondary alkyl halide, 2-bromobutane.

Educational objective:

S_N1 reactions are nucleophilic substitutions that occur in two steps: Formation of a carbocation and nucleophilic attack of the carbocation. Reactivity of alkyl halide substrates depends on the alkyl halides' substitution. Tertiary alkyl halides are the fastest to react because they form a stable carbocation; as substitution decreases, the alkyl halide will react more slowly because it forms a less stable carbocation.

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